30-Minute Student Task

Build Your Own Rule-Based Python Chatbot — Solved

Based on the requirements and target flow in the provided task sheet.

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1. Task Summary

The task is to build a simple rule-based conversational chatbot that continuously interacts with the user, recognizes different messages, gives appropriate responses, and exits when the user types “bye” or “exit.” The required flow is: INPUT → PROCESS → RESPOND → REPEAT → EXIT.

2. Requirement Checklist

Requirement	How it is satisfied
Ask the user's name before the main conversation.	Implemented with input() before the loop.
Use while True.	Used to keep the conversation running.
Use input() to receive messages.	Used for the user's name and every message.
Clean input using .lower().strip().	Applied to every conversation message.
Recognize at least two greetings.	Recognizes hello and hi, including variations using in.
Answer at least three questions/topics.	Covers Python, weather, help, and time-related topics.
Use if / elif / else.	The chatbot uses an if/elif/else decision structure.
Use break for bye or exit.	The loop breaks when the cleaned input is bye or exit.
Use the user's name in at least one response.	The greeting and goodbye use the user's name.
Useful response for unknown messages using else.	Unknown input receives a helpful fallback response.
Challenge: use in operator for flexible greetings.	Recognizes phrases such as 'hello bot' and 'hey hello'.

3. Complete Python Solution

```
print("Welcome to the Rule-Based Python Chatbot!")
name = input("What is your name? ").strip()
print(f"Nice to meet you, {name}!")
print("You can say hello, ask about Python, ask about the weather,")
print("ask for help, or type 'bye' or 'exit' to leave.")
while True:
message = input("You: ").lower().strip()
if message == "bye" or message == "exit":
print(f"Goodbye, {name}! Have a great day.")
break
elif "hello" in message or "hi" in message:
print(f"Hello, {name}! How can I help you?")
elif "python" in message:
print("Python is a beginner-friendly programming language used for web development,
automation, data analysis, and AI.")
elif "weather" in message:
print("I cannot check live weather, but you can use a weather app or website for the current
forecast.")
elif "help" in message:
print("You can greet me, ask about Python, ask about the weather, or type 'bye' or 'exit' to end
the chat.")
elif "time" in message:
print("I cannot access your local clock in this simple rule-based chatbot.")
else:
print("Sorry, I do not understand that yet. Try saying hello, asking about Python or weather,
asking for help, or typing 'bye' to exit.")
```

4. How the Chatbot Works

INPUT: The program first asks the user for a name, then repeatedly uses input() to receive messages.

PROCESS: Each message is cleaned with .lower().strip(), so inputs such as HELLO and hello are treated consistently.

RESPOND: The if/elif/else statements compare the cleaned message with known rules and print the appropriate response.

REPEAT: while True keeps the conversation running after each response.

EXIT: If the user enters bye or exit, the chatbot prints a goodbye message and uses break to leave the loop.

5. Test Cases and Expected Results

Test Input	Expected Result
Name Geethika	Nice to meet you, K. Bhargava Ram!
Greeting hello	Hello, K. Bhargava Ram! How can I help you?
Uppercase greeting HELLO	Hello, K. Bhargava Ram! How can I help you?
Second greeting hi	Hello, K. Bhargava Ram! How can I help you?
Flexible greeting hello bot	Hello, K. Bhargava Ram! How can I help you?
Flexible greeting hey hello	Hello, K. Bhargava Ram! How can I help you?
Question/topic What is Python?	A short explanation of Python is displayed.

Test Input	Expected Result
Question/topic Tell me the weather	The chatbot explains that it cannot check live weather.
Help help	The chatbot lists the supported commands/topics.
Unknown message I like pizza	A useful fallback message is displayed.
Exit bye	A goodbye message is displayed and the loop ends.
Exit exit	A goodbye message is displayed and the loop ends.

6. Sample Demonstration

```
Welcome to the Rule-Based Python Chatbot!

What is your name? K. Bhargava Ram

Nice to meet you, K. Bhargava Ram!

You: HELLO

Hello, K. Bhargava Ram! How can I help you?

You: What is Python?

Python is a beginner-friendly programming language used for web development, automation, data analysis,
and AI.

You: hello bot

Hello, K. Bhargava Ram! How can I help you?

You: I like pizza

Sorry, I do not understand that yet. Try saying hello, asking about Python or weather, asking for help,
or typing 'bye' to exit.

You: bye

Goodbye, K. Bhargava Ram! Have a great day.
```

7. Why This Solution Meets the Task

This solution follows the task sheet rather than relying on a copied chatbot example. It uses the requested conversational loop, input cleaning, multiple greetings, multiple topics, if/elif/else decision making, an exit condition with break, the user's name, and a useful unknown-message response. It also implements the challenge by using the in operator so greeting variations such as “hello bot” and “hey hello” are recognized.

8. Final Target Flow

INPUT → PROCESS → RESPOND → REPEAT → EXIT

The task sheet specifically identifies this as the target flow and asks the student to test greetings, uppercase input, questions, an unknown message, and bye/exit before preparing the demonstration.